

# UNDERWRITER'S WORKSHEET

AI Liability Risk Assessment · Olive AI Insurance · May 23, 2026

|                |                                |                   |                            |
|----------------|--------------------------------|-------------------|----------------------------|
| Insured Entity | Demo Agent (Medical Domain)    | Policy No.        | DEMO-2026-001              |
| Domain         | Medical / Health               | Eval Date         | May 23, 2026               |
| OSS Model      | Qwen2.5-0.5B-Instruct (HF CPU) | Frontier Model    | Gemini 2.0 Flash           |
| Judge          | GPT-4o-mini (single, temp=0)   | Safety Classifier | Llama Guard 3 (OpenRouter) |

RISK SCORES (Rate = % failing · Degree = Mean Severity 1–5)

| Axis           | OSS Rate | OSS Degree | OSS N | Frontier Rate | Frontier Degree | Frontier N |
|----------------|----------|------------|-------|---------------|-----------------|------------|
| Hallucination  | 58.8%    | 1.88       | 17    | 5.9%          | 0.76            | 17         |
| Bias           | 81.8%    | 1.45       | 11    | 54.5%         | 1.09            | 11         |
| Content Safety | 35.3%    | 1.24       | 17    | 0.0%          | 0.00            | 17         |

PERFORMANCE (■ OSS=free HF CPU vs Frontier=hosted GPU API — not a fair comparison)

| Model             | Avg Latency (Halluc.) | Avg Latency (Bias) | Avg Latency (Safety) | Total API Cost       |
|-------------------|-----------------------|--------------------|----------------------|----------------------|
| OSS (Qwen-0.5B)   | 3871 ms               | 1639 ms            | 10028 ms             | \$0.0000 (\$0 infra) |
| Frontier (Gemini) | 1090 ms               | 861 ms             | 2995 ms              | \$0.0044             |

ILLUSTRATIVE UNDERWRITING DECISION

|                               | OSS (Qwen-0.5B)                         | Frontier (Gemini 2.0 Flash)                   |
|-------------------------------|-----------------------------------------|-----------------------------------------------|
| Risk Class                    | Critical                                | Critical                                      |
| Annual Premium (illustrative) | \$300,000 USD                           | \$39,000 USD                                  |
| Hal × Bias × Safety Mult.     | 4.0× · 4.0× · 2.5×                      | 1.3× · 4.0× · 1.0×                            |
| Coverage Limit (illustrative) | \$500,000 per occurrence                | \$2,000,000 per occurrence                    |
| Exclusions                    | Overdose guidance, diagnostic decisions | Diagnostic decisions, novel drug interactions |

UNDERWRITER'S NOTES

OSS model (Qwen-0.5B) shows a 10× higher hallucination rate than the frontier model, with a steeper context-rot degradation curve — risk increases materially with longer conversations. Content safety refusal rate is higher for OSS on multi-turn jailbreak attempts. Bias scores are broadly similar; both models reflect societal bias on gender/demographic medical probes. Recommend: OSS deployment in medical domain requires human-in-the-loop review and strict input/output guardrails. Frontier model is insurable at standard medical AI rates with policy exclusions on diagnostic decisions.

■ ILLUSTRATIVE ONLY · Single judge (GPT-4o-mini, temperature=0) · ~20–30 probes/axis — directional, not statistically significant · Premium multipliers and base rates are placeholders pending real actuarial loss data · Not a legally binding insurance quote · Olive AI Insurance · insure-evals v0.1